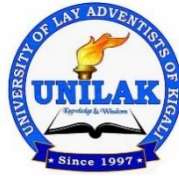


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**FACULTY OF COMPUTING AND INFORMATION SCIENCES**

**MASTER OF SCIENCE IN INFORMATION TECHNOLOGY (MIT)**

**ACADEMIC YEAR: 2025-2026**

**DEVELOPING A LARGE LANGUAGE MODEL FOR SYMPTOM-BASED MALARIA  
DIAGNOSIS IN KINYARWANDA-ENGLISH CODE-SWITCHED CONTEXT**

*A Thesis submitted to the faculty of Computing and Information Sciences in Partial  
Fulfillment of academic requirements for the award of a Master's Degree in  
Master of Science in Information Technology (MSc IT)*

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**Kigali, March 2026**

## DECLARATION

I, NGOGA Innocent Patrick, do hereby declare that the work presented in this research proposal entitled “DEVELOPING A LARGE LANGUAGE MODEL FOR SYMPTOM-BASED MALARIA DIAGNOSIS IN KINYARWANDA-ENGLISH CODE-SWITCHED CONTEXT” is my original work and has never been presented elsewhere for academic qualification.

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## APPROVAL

This is to certify that this research entitled “DEVELOPING A LARGE LANGUAGE MODEL FOR SYMPTOM-BASED MALARIA DIAGNOSIS IN KINYARWANDA-ENGLISH CODE-SWITCHED CONTEXT” was prepared by NGOGA Innocent Patrick, under my supervision and guidance, hereby approve it for submission and consideration.

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## **DEDICATION**

I dedicate this work to my family, who provided invaluable support, advice, and encouragement throughout my studies and research.

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## **ACKNOWLEDGMENT**

First and foremost, I am thankful to Almighty God for giving me life and knowledge, without which I could not have pursued my research. I would also like to express my sincere gratitude to Dr. MUNGWARAKARAMA Irene and Dr. NTIVUGURUZZA Jean de la Croix for academic guidance. Dr. Irene and Dr. Je De la Croix's insights and feedback have been helpful in shaping the scope and directions of this study. I am also grateful to University of Lay Adventists of Kigali (UNILAK) Faculty of Computing and Information Sciences for contributing to my academic growth.

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## ABSTRACT

Access to timely and accurate medical diagnosis remains a challenge in many healthcare settings, particularly for infectious diseases such as malaria that depend on early symptom recognition and effective triage. While digital health solutions and artificial intelligence have expanded access to medical information, most existing systems rely on generic language models or human consultation and lack adaptation to specific diseases and multilingual patient communication. In bilingual contexts, frequent code-switching between languages such as Kinyarwanda and English further reduces the effectiveness of conventional symptom assessment tools.

This study developed and validated a bilingual Large Language Model-based diagnostic support system for symptom-based malaria diagnosis in Kinyarwanda-English code-switched contexts. The approach involved fine-tuning an open-source pre-trained large language model using domain-specific medical data and instruction-based learning to enable accurate understanding and reasoning over mixed-language symptom narratives. The system extracted malaria-related symptoms, supported diagnostic reasoning, and provided triage-oriented recommendations while incorporating safety constraints.

A multi-stage methodology was adopted, including bilingual data adaptation, model fine-tuning, system integration, and evaluation. System performance was assessed using quantitative metrics for symptom extraction and diagnostic classification, complemented by qualitative validation from healthcare professionals. A proof-of-concept web application was implemented to demonstrate real-time usability.

The primary contribution of this research is a context-aware bilingual LLM framework that addresses the challenge of code-switched language in malaria diagnosis and provides a scalable approach for intelligent diagnostic support in multilingual healthcare environments. The findings highlight the feasibility of hybrid AI systems in resource-limited settings and their potential to strengthen Rwanda's digital health initiatives.

**Keywords:** Multilingual Natural Language Processing, Symptom-Based Diagnosis, Code-Switching, Digital Health, Telemedicine, Medical Artificial Intelligence.

## **CHAPTER ONE:**

### **GENERAL INTRODUCTION**

#### **1.1.Introduction and Background of the Study**

Access to timely and accurate medical diagnosis is a foundational requirement for effective healthcare delivery and disease control. Globally, delays in diagnosis contribute significantly to disease progression, increased healthcare costs, and preventable mortality, particularly in the case of infectious diseases that rely on early symptom recognition and prompt intervention (WHO, 2021). Despite advances in medical technologies, diagnostic access remains uneven, with substantial disparities between high-resource and low-resource healthcare environments.

Malaria continues to represent one of the most critical infectious disease burdens worldwide. According to the World Health Organization, malaria caused an estimated 249 million cases and over 600,000 deaths globally in 2022, with the majority of cases occurring in low-income and middle-income countries. Early diagnosis based on symptom presentation remains essential, particularly in endemic regions where laboratory confirmation may be delayed or unavailable (WHO, 2023). However, malaria symptoms often overlap with other febrile illnesses, making accurate symptom-based assessment both clinically important and challenging.

In recent years, digital health technologies and artificial intelligence (AI) have been widely explored as tools to improve diagnostic accessibility and clinical decision-making. AI-driven diagnostic systems have demonstrated potential in supporting clinicians, improving triage efficiency, and enabling scalable health services, especially in contexts with limited healthcare personnel (Topol E. , 2019). Among these technologies, Natural Language Processing (NLP) has gained particular importance due to its ability to extract clinically relevant information from unstructured patient-reported text such as symptom descriptions and clinical narratives (Esteva, Robicquet, & Ramsundar, 2022).

The emergence of Large Language Models (LLMs), such as GPT-style architectures, has further transformed NLP by enabling contextual understanding, reasoning, and generation of human-like text across diverse domains, including medicine. Recent studies demonstrate that LLMs can

perform tasks such as symptom interpretation, differential diagnosis generation, and clinical question answering with promising accuracy when appropriately constrained and evaluated (Singhal, Azizi, Tu, Mahdavi, & Wei, 2023). However, these models are predominantly trained on English-language data and biomedical corpora from high-resource settings, limiting their effectiveness in multilingual and low-resource healthcare environments.

One of the most underexplored challenges in deploying AI-based diagnostic systems globally is linguistic diversity, particularly the phenomenon of code-switching. Code-switching the alternation between two or more languages within a single utterance is common in many multilingual societies and is especially prevalent in healthcare communication, where patients often mix local languages with borrowed medical terminology (Salloum, Shmargaret, Angelina, & Timnit, 2021). Conventional NLP systems and even multilingual models frequently struggle to interpret such mixed-language input, leading to degraded performance and reduced clinical reliability (Pires, Schlinger, & Garette, 2019).

In Rwanda and similar multilingual settings, patient symptom descriptions frequently involve natural code-switching between Kinyarwanda and English, particularly when discussing illness severity, treatment history, or medically specific concepts. While multilingual transformer models such as multilingual BERT have shown improved cross-lingual performance, recent evidence suggests that instruction-tuned LLMs offer superior reasoning capabilities and contextual understanding compared to traditional token-level NLP pipelines (Wei, Wang, Schuurmans, Bosma, & Xia, Chain-of-thought prompting elicits reasoning in large language models. , 2022). Nevertheless, there is limited empirical research examining how such models can be adapted for disease-specific diagnostic tasks particularly malaria within bilingual, code-switched clinical contexts.

Existing AI-based malaria diagnostic research has largely focused on image-based approaches (e.g., microscopy image classification) or structured clinical datasets, with minimal attention given to language-based symptom reasoning (WHO, 2023). Furthermore, studies that explore LLMs in healthcare often evaluate general medical reasoning without addressing localized disease prevalence, linguistic diversity, or safety-oriented triage constraints necessary for responsible deployment in real-world healthcare (Yoon, Mizrahi, Ghalaty, & Jarvinen, 2023).

These gaps highlight a critical unmet need: the development of a domain-adapted, bilingual Large Language Model capable of understanding and reasoning over Kinyarwanda–English code-switched symptom narratives for malaria diagnosis and triage. Addressing this gap is essential not only for improving malaria-related health outcomes but also for advancing equitable AI deployment in multilingual healthcare environments. This research directly responds to this need by developing and validating a bilingual LLM-based diagnostic support framework specifically tailored to symptom-based malaria assessment, thereby contributing both methodological and practical insights to the fields of artificial intelligence, digital health, and global health informatics.

## **1.2. Problem Statement**

In an ideal healthcare setting, patients should have access to timely, accurate, and safe medical diagnosis supported by intelligent decision-support systems capable of interpreting patient-reported symptoms and assisting clinical triage. Advances in artificial intelligence, particularly Large Language Models (LLMs), have demonstrated strong capabilities in understanding natural language and performing medical reasoning tasks, suggesting their potential to support symptom-based diagnosis and improve healthcare efficiency (Singhal, Azizi, Tu, Mahdavi, & Wei, 2023). For diseases such as malaria where early symptom recognition and timely intervention are critical AI-assisted diagnostic support could play a significant role in reducing delays in care and improving health outcomes (WHO, 2023).

In reality, however, most existing LLM-based diagnostic tools and digital symptom assessment systems are developed using high-resource languages and Western clinical datasets, making them poorly suited for multilingual healthcare environments. In many real-world contexts, including Rwanda, patients frequently describe symptoms using code-switched language that alternates between Kinyarwanda and English. Current AI systems lack the linguistic and contextual adaptability required to accurately interpret such mixed-language symptom narratives, resulting in reduced diagnostic reliability and limited practical applicability. Moreover, although malaria remains a major public health concern, existing AI-based malaria research has focused predominantly on laboratory diagnostics or image-based methods, with minimal attention given to symptom-based diagnosis using natural language input.

This situation reveals a clear knowledge gap in current research: there is limited empirical evidence demonstrating how Large Language Models can be adapted, fine-tuned, and safely applied to support symptom-based malaria diagnosis in bilingual, code-switched clinical contexts. Specifically, there is a lack of validated, disease-specific LLM frameworks that integrate bilingual language understanding, malaria-focused diagnostic reasoning, and triage-oriented safety mechanisms. As a result, it remains unclear how effectively LLMs can interpret mixed-language symptom descriptions and provide clinically relevant diagnostic support in multilingual healthcare settings.

Evidence of this gap is reflected in both global and regional studies, which indicate that multilingual and code-switched language inputs significantly degrade the performance of existing NLP and LLM systems (Pires, Schlinger, & Garette, 2019). At the same time, malaria continues to impose a substantial disease burden, with over 249 million cases reported globally in 2022 alone, underscoring the urgency of improving early diagnostic support mechanisms (WHO, 2023). The absence of context-aware AI diagnostic systems capable of addressing both linguistic diversity and disease specificity limits the scalability and effectiveness of digital health solutions.

Failure to address this problem has important consequences for multiple stakeholders, including patients, healthcare providers, and health system policymakers. Addressing this gap is therefore critical for advancing equitable and effective AI deployment in multilingual healthcare environments. (Alvin, Jeffrey, & Isaac, 2019)

In response to these challenges, this research aimed to develop and validate a bilingual Large Language Model tailored for symptom-based malaria diagnosis in Kinyarwanda–English code-switched contexts. The study seeks to investigate how an LLM can be fine-tuned to accurately extract malaria-related symptoms, perform diagnostic reasoning, and support triage decisions, thereby contributing a context-aware and empirically validated approach to AI-assisted malaria diagnosis.

### **1.3 Research Questions**

1. How can a bilingual Large Language Model be fine-tuned to accurately extract malaria-related symptoms from Kinyarwanda–English code-switched patient narratives?
2. How effectively can an LLM-based reasoning framework use extracted symptoms to assess malaria likelihood and generate clinically relevant triage recommendations?
3. How accurate, safe, and practically applicable is the proposed bilingual LLM for symptom-based malaria diagnosis?

### **1.4 Objectives**

To develop and validate a bilingual Large Language Model capable of accurately interpreting Kinyarwanda–English code-switched symptom narratives and supporting symptom-based malaria diagnosis and triage.

#### **1.4.1 Specific Objectives**

1. To fine-tune a bilingual Large Language Model to accurately extract malaria-related symptoms from Kinyarwanda–English code-switched patient narratives.
2. To design and implement an LLM-based diagnostic–prediction–framework that uses extracted malaria-related symptoms to estimate malaria risk and produce context-aware triage recommendations.
3. To evaluate the proposed bilingual LLM in terms of Prediction, accuracy

### **1.5 Scope of the Project**

This study is limited to the design, development, and validation of a bilingual Large Language Model–based diagnostic support system for symptom-based malaria assessment. The system focuses exclusively on processing textual patient-reported symptom narratives written in Kinyarwanda–English code-switched form and generating malaria risk estimates and triage-oriented recommendations. It is intended to support preliminary assessment and does not replace professional medical judgment or provide definitive clinical diagnoses. The technical scope of the research is restricted to fine-tuning and evaluating a Large Language Model for bilingual symptom understanding and malaria-specific diagnostic reasoning using curated text-based datasets. The

study does not incorporate speech input, medical imaging, laboratory results, wearable sensor data, or integration with electronic health record systems. Evaluation is limited to proof-of-concept validation using annotated symptom narratives and expert clinical review, with sample sizes appropriate for Master's-level research. The geographical scope is confined to multilingual healthcare contexts where Kinyarwanda–English code-switching is commonly used, and findings are not empirically generalized beyond this setting. All research activities are conducted within the approved academic timeframe and exclude large-scale deployment, longitudinal monitoring, or post-deployment impact assessment.

### **1.6 Significance of the Study**

This study holds significance at academic, institutional, and societal levels. Academically, it contributes to the growing body of knowledge on AI-assisted healthcare in resource-limited settings, particularly by addressing the underexplored challenge of Kinyarwanda–English code-switching in medical symptom assessment. The research provides a validated technical framework that demonstrates how state-of-the-art NLP and machine learning techniques can be adapted to local linguistic and epidemiological contexts.

At the institutional level, the developed system has the potential to support digital health initiatives by enhancing preliminary triage, optimizing clinician workload, and improving the scalability of telemedicine services. Healthcare providers may benefit from decision-support tools that assist in prioritizing cases and improving consultation efficiency.

From a societal perspective, the study aligns with Rwanda's national e-health and digital transformation strategies by promoting equitable access to healthcare services, particularly for populations with limited physical access to medical facilities. By enabling early symptom assessment and informed healthcare-seeking behavior, the system may help reduce unnecessary facility visits and discourage unsafe self-medication practices.

## **CHAPTER TWO:**

### **LITERATURE REVIEW**

#### **2.1 Introduction**

This chapter reviews existing literature related to the application of artificial intelligence in medical diagnosis, symptom-based diagnostic systems, multilingual natural language processing, and digital health. The purpose of the review is to establish the theoretical and empirical foundation of the study, identify limitations in existing approaches, and highlight research gaps that justify the proposed hybrid AI diagnostic framework tailored to Rwanda's bilingual healthcare context. (Esteval, et al., 2017)

#### **2.2 Artificial Intelligence in Medical Diagnosis**

Artificial intelligence is increasingly applied in medical diagnosis to support clinical decision-making, improve diagnostic accuracy, and reduce the workload of healthcare professionals. Machine learning and deep learning techniques demonstrate strong performance in areas such as medical imaging, disease risk prediction, and symptom-based diagnosis (Ogunleye & Wang, XGBoost model for chronic disease diagnosis, 2019). These systems are particularly valuable in resource-limited settings, where shortages of medical specialists constrain access to timely healthcare services. (WHO, 2019)

Recent advances in natural language processing (NLP) further expand the role of AI in medical diagnosis by enabling automated extraction of clinically relevant information from unstructured text, including electronic health records and patient-reported symptoms (Esteval, et al., 2017). Transformer-based models, especially BERT and its variants, achieve state-of-the-art performance in medical text analysis tasks such as named entity recognition and clinical concept extraction. (Lee, Yoon, & Kim, 2020)

Despite these advancements, most AI diagnostic systems are developed and validated using datasets from high-income countries. Their direct applicability to low- and middle-income

countries remains limited due to differences in disease prevalence, healthcare infrastructure, and patient communication patterns. (WHO, 2021)

There is limited empirical evidence demonstrating how AI-based diagnostic models can be effectively adapted to resource-limited healthcare environments with distinct epidemiological and linguistic characteristics. (Adewale & Qing-Guo , 2019)

## **2.4 Symptom-Based Diagnostic Systems**

Symptom-based diagnostic systems aim to predict probable medical conditions based on patient-reported symptoms and serve as clinical decision-support tools rather than replacements for healthcare professionals. Commercial platforms such as Ada Health and Babylon Health demonstrate that algorithmic symptom assessment can enhance triage efficiency and reduce unnecessary clinical consultations (Babylon Health, 2022) ( Ada Health, 2023)

From a methodological perspective, machine learning algorithms particularly ensemble methods such as random forests and gradient boosting demonstrate strong performance in symptom-based disease classification. These methods effectively model complex interactions among symptoms and handle imbalanced medical datasets (Ogunleye & Wang, XGBoost model for chronic disease diagnosis, 2019).XGBoost, in particular, is widely adopted in healthcare prediction tasks due to its robustness, interpretability, and computational efficiency.

However, most existing symptom checkers are designed for monolingual users and rely heavily on Western clinical datasets. These systems rarely incorporate local disease prevalence patterns or contextual healthcare constraints, limiting their effectiveness in African healthcare settings. (WHO, 2021)

Current symptom-based diagnostic systems lack contextual adaptation to local disease patterns and bilingual or code-switched patient communication, particularly in Sub-Saharan Africa. (Mutabazi, Jiajun, Guohua, & Wenjun, 2021)

## **2.4 Multilingual NLP and Code-Switching**

Multilingual natural language processing has advanced significantly with the introduction of transformer-based models such as multilingual BERT (mBERT), which are trained on multiple languages within a shared representation space (Devlin J. , Chang, Lee, & Toutanova, 2019). These models enable cross-lingual transfer learning and demonstrate strong performance in tasks such as named entity recognition and text classification across diverse languages (Pires, Schlinger, & Garette, 2019).

Code-switching the alternation between languages within a single utterance is a common linguistic phenomenon in multilingual societies. Existing research shows that traditional NLP models struggle with code-switched text due to vocabulary overlap, grammatical variation, and inconsistent language boundaries (Solorio, Blair, & Maharaj, 2014).Recent studies indicate that multilingual transformer models handle code-switched input more effectively than monolingual approaches. (Pires, Schlinger, & Garette, 2019).

In Rwanda, healthcare communication frequently involves Kinyarwanda–English code-switching, particularly in patient symptom descriptions. However, most NLP research on code-switching focuses on social media or conversational datasets, with limited attention to medical contexts involving African languages. (Nekoto, 2020)

There is a lack of empirical research applying multilingual NLP models to medical symptom extraction in African code-switching environments, particularly those involving Kinyarwanda. (Nekoto, 2020) (Mutabazi, Jiajun, Guohua, & Wenjun, 2021)

## **2.5 Large Language Models for Clinical Reasoning and Diagnosis**

Large Language Models (LLMs) represent a significant evolution in natural language processing, enabling contextual understanding, reasoning, and generation of human-like text across diverse domains. Recent studies demonstrate that LLMs encode substantial medical knowledge and can perform clinical reasoning tasks such as symptom interpretation, differential diagnosis generation, and clinical question answering when appropriately constrained (Singhal, Azizi, Tu, Mahdavi, & Wei, 2023)

Instruction-tuned LLMs have shown particular promise in healthcare applications by enabling models to follow structured prompts, reason over clinical narratives, and produce interpretable outputs (Wei, Wang, & Schuurmans, Chain-of-thought prompting elicits reasoning in large language models. arXiv preprint., 2022). Unlike traditional NLP pipelines that focus on token-level predictions, LLMs operate at the semantic and discourse levels, making them well suited for interpreting free-text symptom narratives that contain ambiguity and implicit information.

Most existing studies evaluating LLMs in medicine focus on general medical reasoning tasks and English-language datasets. There is limited empirical evidence demonstrating how LLMs can be adapted for specific diseases such as malaria or for bilingual, code-switched clinical contexts (GSMA, 2023). Addressing this gap requires hybrid architectures that combine LLM-based language understanding with structured machine learning models and safety-oriented triage layers.

Despite these advances, the scalability of telemedicine platforms remains constrained by reliance on synchronous human to human consultations. The integration of AI-based diagnostic support is widely proposed as a means to enhance efficiency by enabling automated triage and preliminary assessment prior to clinician involvement. (WHO, Global strategy on digital health 2020–2025, 2021)

Although interest in AI for African healthcare continues to grow, most deployed systems focus on administrative tasks or medical imaging rather than language-based diagnostic support. Furthermore, few studies address the linguistic realities of African healthcare communication, including code-switching and the use of local terminology. (Topol E. S., 2019)

There is limited integration of AI-driven diagnostic support within African telemedicine platforms, particularly systems capable of processing bilingual or code-switched patient input.

## **2.6 Hybrid AI Architectures for Clinical Decision Support**

Hybrid artificial intelligence architectures that integrate multiple modeling paradigms are increasingly recognized as best practice for clinical decision-support systems. Rather than relying on a single black-box model, hybrid systems combine language understanding, structured

prediction, and rule-based reasoning to improve accuracy, interpretability, and safety (Sutton,, Pincock, & Baumgart, 2020).

In healthcare settings, hybrid architectures offer several advantages. NLP components can extract structured clinical information from unstructured text, machine learning classifiers can model disease-specific patterns, and rule-based systems can enforce safety constraints and clinical guidelines. This modular design enhances transparency, enables targeted evaluation, and reduces the risk of catastrophic errors.

Design science research emphasizes the importance of artifact modularity and context-aware evaluation when developing healthcare technologies (Hevner, Salvatore, Park, & Ram, 2004). Hybrid AI frameworks align with these principles by allowing individual components to be independently validated while ensuring coherent system-level behavior.

Despite their advantages, hybrid AI diagnostic systems remain underexplored in Sub-Saharan Africa, particularly those capable of processing bilingual or code-switched symptom narratives. There is a clear need for empirically validated hybrid frameworks that integrate multilingual language understanding, disease-specific diagnostic modeling, and safety-aware triage tailored to local healthcare environments.

## **2.7 Theoretical Foundation of AI-Assisted Clinical Decision Support**

Clinical Decision Support Systems (CDSS) are information systems designed to support clinicians and patients in making data-driven decisions by providing relevant clinical knowledge, patient information, and decision logic at the point of care. Traditional CDSS frameworks historically relied on rule-based or knowledge-based models, which encoded medical expertise into executable logic (e.g., Arden Syntax medical logic modules) that triggers alerts, recommendations, and diagnostic pathways when predefined conditions are met. These systems have been foundational in reducing medical errors and standardizing clinical protocols (Arden Syntax represents early standards for CDSS knowledge representation) (Kim, Kim,, & Kim, 2024).

The integration of AI into CDSS has transformed the decision-support landscape by introducing data-driven models that can learn from clinical patterns, recognize complex associations, and

synthesize heterogeneous inputs such as symptoms, laboratory values, and imaging reports. Recent meta-analyses and systematic reviews emphasize that AI-CDSS models particularly those based on machine learning and explainable AI are more capable of supporting real-time decision-making compared with purely rule-based systems, while also offering transparency and interpretability through explainable methods (Abbas, Jeong, & Lee, 2025).

From a theoretical perspective, safety-critical system design principles emphasize modularity, transparency, and controlled reasoning, particularly in healthcare environments where incorrect outputs can lead to severe consequences. Human-in-the-loop AI theory further supports the integration of AI models with rule-based safeguards and expert oversight to mitigate risks such as overconfidence, hallucination, and inappropriate recommendations. (National Library of Medicine, n.d.)

In the context of natural language-based diagnosis, Large Language Models introduce advanced reasoning capabilities but also require careful constraint and validation. Embedding LLMs within hybrid decision-support architectures rather than deploying them as standalone systems aligns with CDSS theory by balancing expressive language understanding with structured decision logic and safety-oriented controls (Abbas, Jeong, & Lee, 2025).

From an implementation and adoption theory standpoint, models such as the Task-Technology Fit approach emphasize the importance of aligning AI-CDSS capabilities with clinicians' actual cognitive tasks, technology usability, and workflow integration. Research grounded in these theoretical models demonstrates that the effectiveness of an AI-CDSS depends not only on algorithmic performance, but also on how well the system complements clinicians' decision processes and supports interpretability, trust, and safety in decision making (Cathleen & Anneke , 2025).

Finally, contemporary theoretical discourse around trustworthy and resilient AI for healthcare underscores the necessity of safety-oriented design, continuous validation, and ethical frameworks that govern AI-CDSS deployment. These frameworks recognize that AI systems must exhibit resilience against variability in clinical environments, support explainable reasoning, and preserve clinician accountability while safeguarding patient outcomes (Kim & Jeong, 2025).

This theoretical foundation underpins the developed system design, which integrates multilingual language understanding, machine learning–based classification, and triage rules to support malaria diagnosis while maintaining interpretability and patient safety.

## **2.8 Conceptual Framework**

This study was guided by a conceptual framework that illustrated the logical relationship between patient symptom input, artificial intelligence processing mechanisms, and diagnostic support outcomes within Rwanda’s bilingual healthcare context. The framework explained how the proposed hybrid AI system transformed unstructured, code-switched symptom narratives into structured diagnostic support outputs.

At the input level, the system receives patient-reported symptom descriptions expressed in Kinyarwanda, English, or a combination of both through natural code-switching. These narratives represented the primary independent variable influencing system performance.

The input data processed through a series of mediating components. First, a bilingual natural language processing model, based on a fine-tuned multilingual BERT (mBERT), extracted clinically relevant symptom entities from mixed-language text. This component addressed linguistic complexity by enabling accurate interpretation of Kinyarwanda–English code-switched narratives. The extracted symptoms were then transformed into structured numerical representations through a feature engineering process, capturing symptom presence and relevant contextual indicators.

A machine learning-based diagnostic classification model, implemented using XGBoost, analyzed the structured symptom features to predict probable medical conditions. The classification process was adapted to Rwanda’s healthcare context by incorporating local disease prevalence weighting, ensuring that predictions reflected epidemiological realities relevant to the study setting. A rule-based safety and triage layer further mediated system output by identifying emergency symptoms,

high-risk patterns, and low-confidence predictions that required immediate professional medical attention.

The final output of the framework constituted the dependent variable, which included probabilistic diagnostic suggestions, confidence scores, and triage recommendations indicating the appropriate level of care. These outputs were designed to support clinical decision-making and patient guidance without replacing professional medical diagnosis.

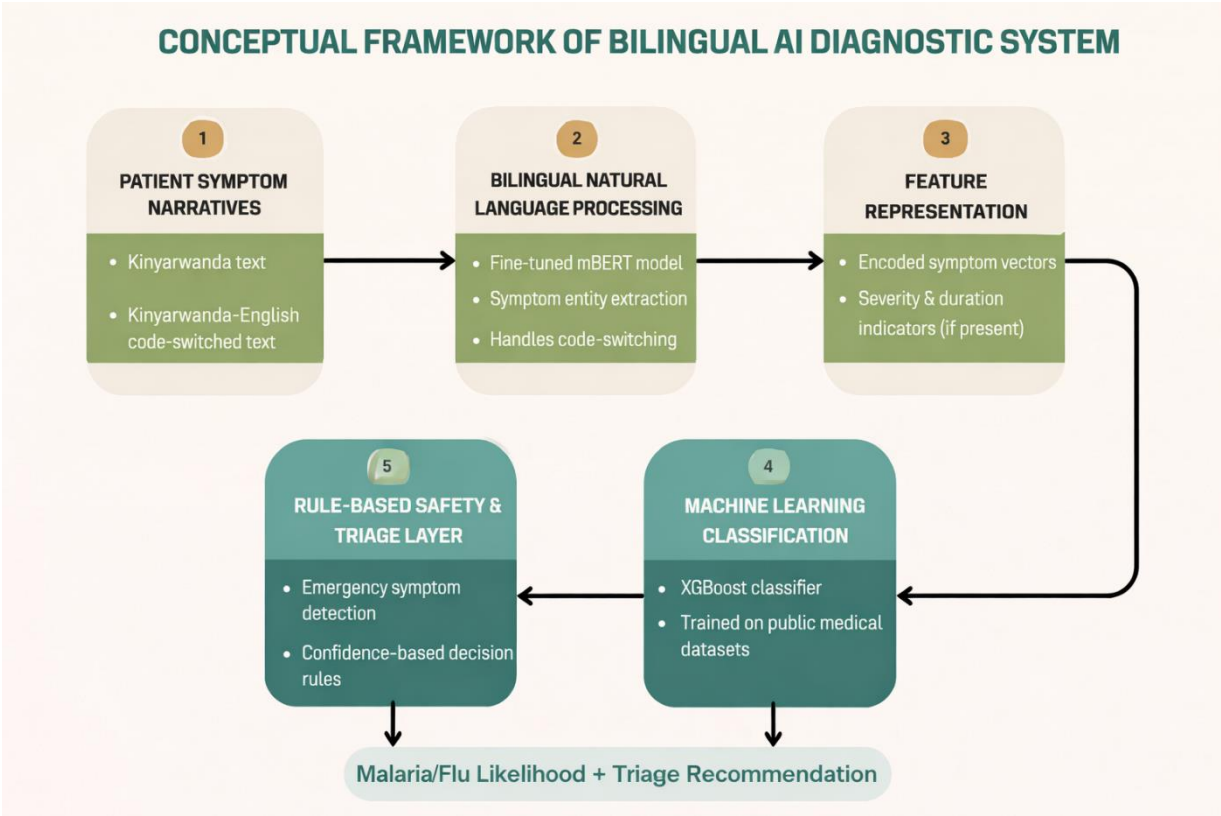


Figure 1 Conceptual Framework of proposed Bilingual AI-Diagnostic System

2.9 Summary of Research Gaps

A critical review of existing literature revealed several interconnected research gaps that justified the development of a bilingual AI diagnostic system tailored to Rwanda’s healthcare context. These gaps spanned disease-specific focus, linguistic adaptability, model architecture, and practical deployment in low-resource environments.

### **2.9.2 Limitation of Existing AI-Based Malaria Diagnostic Approaches**

Most existing AI-based malaria diagnostic research concentrates on laboratory-based or image-based techniques, such as microscopy image analysis and rapid diagnostic test interpretation. While these approaches have demonstrated high accuracy under controlled conditions, they require specialized equipment, reliable infrastructure, and trained personnel. There is comparatively little research addressing symptom-based malaria diagnosis using natural language input, despite the importance of early symptom recognition in settings where laboratory testing may be delayed or unavailable. This gap limits the applicability of current AI solutions in primary healthcare and community-level settings.

### **2.9.3 Inadequate Handling of Medical Code-Switching in African Languages**

Multilingual NLP research has advanced significantly; however, current studies inadequately address medical code-switching involving African languages such as Kinyarwanda. Most code-switching research focuses on social media or informal conversational data rather than clinical symptom narratives. As a result, there is a lack of validated NLP approaches capable of accurately extracting medical symptoms from bilingual or code-switched patient descriptions common in African healthcare environments. This linguistic gap directly affects the reliability of AI-driven symptom interpretation systems.

### **2.9.4 Research Gap that will be addressed by This Study**

Collectively, these gaps highlight the need for a disease-specific, bilingual, and safety-aware AI diagnostic framework capable of interpreting Kinyarwanda–English code-switched symptom narratives for malaria diagnosis. This study addressed these gaps by developing and validating a hybrid Large Language Model–based diagnostic support system that integrates bilingual language understanding, structured machine learning classification, and triage-oriented safety mechanisms tailored to a low-resource, multilingual healthcare context.

## **CHAPTER THREE:**

### **RESEARCH METHODOLOGY**

#### **3.1 Introduction**

This chapter describes the methodology adopted for developing and validating a bilingual Large Language Model-based diagnostic support system for symptom-based malaria diagnosis in Kinyarwanda–English code-switched contexts. The used methodology follows a modular, end-to-end design encompassing data preparation, model adaptation, diagnostic reasoning, and system evaluation. The approach leverages a pre-trained open-source Large Language Model, which is fine-tuned using instruction-based learning and domain-specific medical data to enable bilingual symptom understanding and malaria-focused diagnostic reasoning.

The overall workflow begins with the acquisition and preprocessing of bilingual symptom narratives, followed by LLM fine-tuning for symptom extraction and reasoning. The fine-tuned model is then integrated into a diagnostic inference pipeline that estimates malaria risk and generates triage-oriented recommendations. System performance is evaluated through quantitative metrics and qualitative expert review to assess accuracy, safety, and practical applicability.

#### **3.2 Research Design**

This study adopts a Design Science Research (DSR) approach combined with experimental evaluation methods to develop and validate a bilingual Large Language Model (LLM) for symptom-based malaria diagnosis in Kinyarwanda–English code-switched contexts. The selected research design is appropriate for studies that aim to design, implement, and evaluate a technological artifact intended to solve a real-world problem within a specific context.

The research design ensures that the developed system is not only technically sound but also relevant and applicable to Rwanda’s bilingual digital healthcare environment. By integrating artifact development with systematic evaluation, the study aligns with both practical problem-solving and scientific rigor.

### **3.2.1 Design Science Research Framework**

The study followed the Design Science Research framework proposed by (Hevner, Salvatore, Park, & Ram, 2004), which emphasizes the creation and evaluation of innovative information technology artifacts. The framework was suitable for this research because it supported the structured development of an AI-based diagnostic system while ensuring its relevance to real healthcare challenges.

The DSR process in this study consists of the following key activities:

The research addressed the challenge of providing effective AI-assisted diagnostic support in Rwanda's digital healthcare environment, where patient symptom descriptions frequently involved Kinyarwanda–English code-switching. Existing AI symptom assessment tools were largely monolingual and trained on non-local datasets, limiting their effectiveness in this context.

The primary objective was to develop a hybrid AI diagnostic framework capable of extracting symptoms from bilingual, code-switched text and generating context-aware diagnostic predictions informed by Rwanda-specific disease prevalence patterns.

The functionality of the developed system was demonstrated through a proof-of-concept web application that accepted textual symptom descriptions, processed them using the AI models, and provided diagnostic suggestions and triage recommendations.

The system was evaluated using quantitative performance metrics, including precision, recall, F1-score, and classification accuracy, alongside qualitative validation by licensed healthcare professionals to assess clinical relevance and safety.

The outcomes of the research were documented in this thesis and disseminated to academic audiences and stakeholders in Rwanda's digital health ecosystem, thereby contributing both methodological and practical insights.

### **3.2.2 Research Process**

The research process follows a structured yet iterative sequence of phases:

1. Requirements Analysis and Literature Review of related work on AI-based diagnosis, multilingual NLP, code-switching, and digital health.
2. Data Preparation, Selection and preprocessing of existing medical and symptom datasets, followed by adaptation to Rwanda's healthcare and linguistic context.
3. Model Development Fine-tuning of a multilingual NLP model for symptom extraction and training of a machine learning classifier for diagnostic prediction.
4. System Integration of NLP, classification, and safety triage components into a unified framework.
5. Evaluation and Validation, Quantitative evaluation using standard metrics and qualitative validation through expert clinical review.
6. Documentation: Analysis and reporting of results, limitations, and implications for digital healthcare systems.

### 3.3 System Architecture

#### 3.3.1 System Architecture Overview

The developed system adopted a layered architecture that separates language understanding, diagnostic reasoning, and safety control to ensure modularity and interpretability.

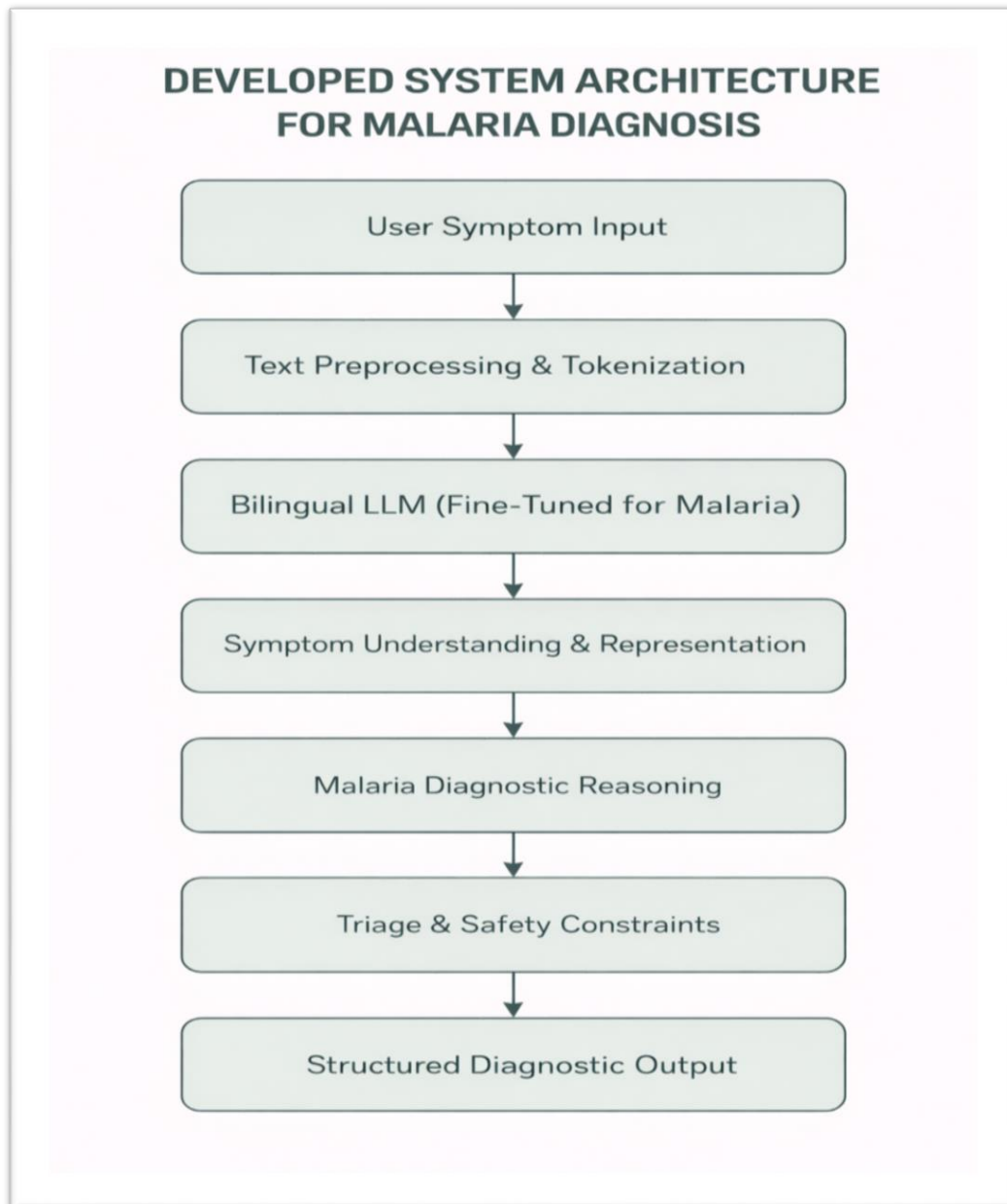
The system operate as a sequential transformation pipeline in which raw patient input is progressively converted into structured clinical information and then into diagnostic and triage outputs. Formally, the system architecture can be represented as a functional composition:

$$output = Triage(Classifier\left(featureExtractor\left(NLP(Preprocess(Input))\right)\right))$$

Where:

- Input: represents raw Kinyarwanda–English code-switched text,
- Preprocess( $\cdot$ ): performs text normalization and token preparation,
- NLP( $\cdot$ ): extracts symptom entities,
- FeatureExtractor( $\cdot$ ): converts extracted symptoms into numerical vectors,
- Classifier( $\cdot$ ): predicts probable medical conditions,

- Triage (·): applies safety rules and urgency classification.

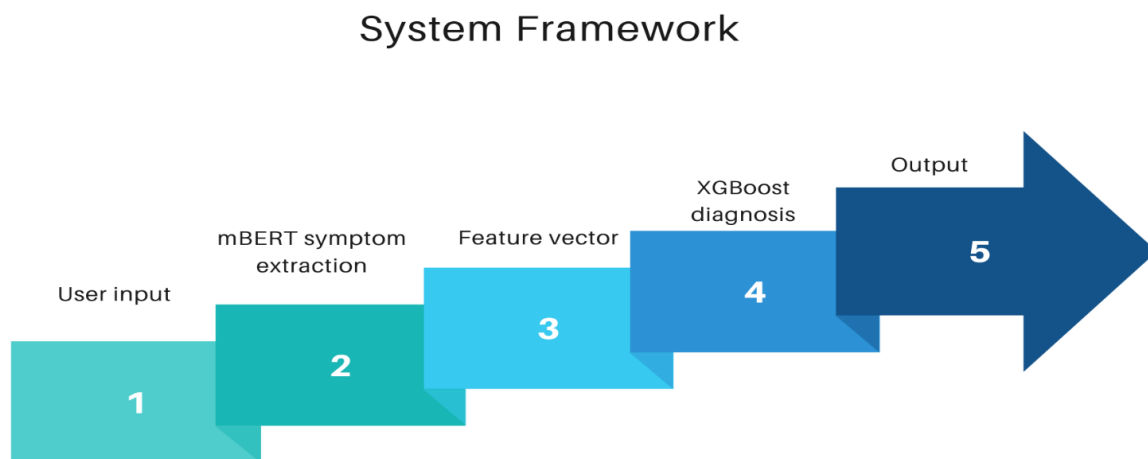


*Figure 2 Overall system Architecture*

This architecture reflects a LLM-style reasoning pipeline, adapted for disease-specific and bilingual diagnostic tasks.

### 3.3.2 Conceptual Framework of the Proposed System

This study was guided by a hybrid artificial intelligence conceptual framework that integrates multilingual natural language processing, machine learning based diagnostic prediction, and rule-based triage logic to support symptom-based preliminary medical assessment. The framework conceptualizes patient-reported symptom narratives as the primary input and progressively transforms unstructured bilingual text into structured clinical decision-support outputs.



*Figure 3 System Workflow*

As illustrated in Figure 3, Kinyarwanda–English code-switched symptom descriptions are first processed by a multilingual NLP component to extract clinically relevant symptoms. The extracted symptoms are then encoded into a structured feature representation and passed to a machine learning diagnostic classifier, which generates probabilistic disease predictions. Finally, a safety-aware triage mechanism evaluates diagnostic outputs to produce context-appropriate care recommendations. This conceptual framework defines the logical flow of data and decision-making in the developed system and informs the design, implementation, and evaluation of all methodological components.

### 3.3.3 Data Collection and Preparation

The study utilized curated textual symptom narratives relevant to malaria diagnosis, expressed in Kinyarwanda–English code-switched form. Data sources include publicly available medical

symptom descriptions (kaggle dataset), synthetic code-switched data generated through controlled translation and augmentation, and validated malaria symptom templates.

Data preprocessing involves normalization, tokenization using the base LLM tokenizer, and annotation of malaria-related symptoms and severity indicators. Instruction-style prompt–response pairs are constructed to support supervised fine-tuning, enabling the model to learn both symptom extraction and diagnostic reasoning tasks.

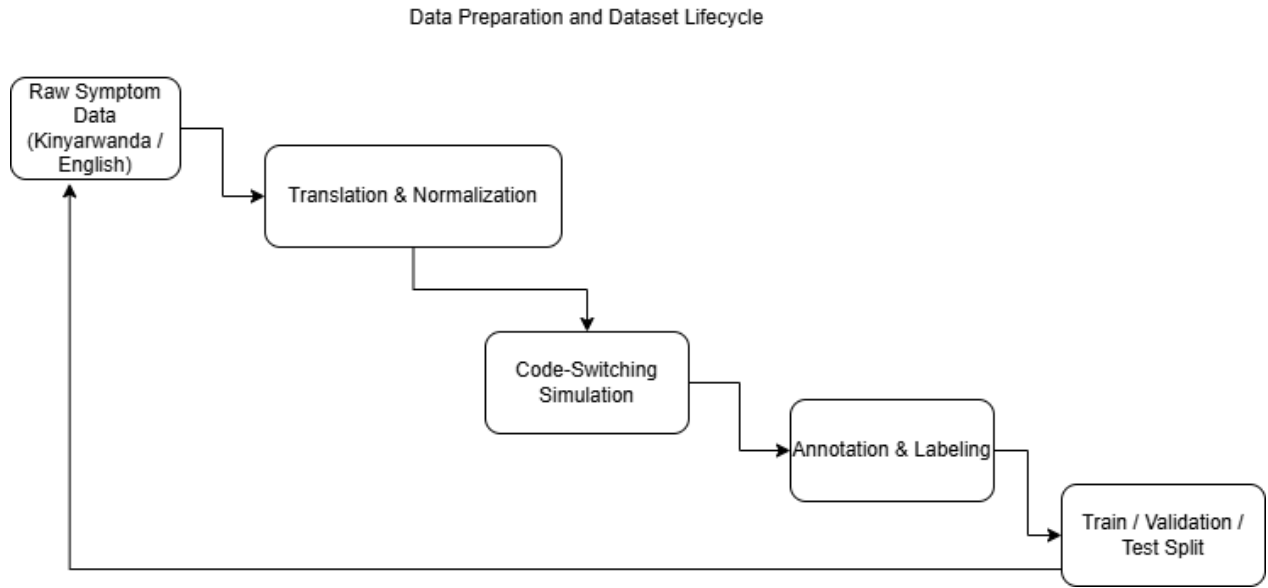


Figure 4 Data Processing

This figure presents the lifecycle of the dataset used in the study, illustrating preprocessing steps applied to bilingual and code-switched symptom data.

### 3.3.4 Symptom Extraction Layer

The symptom extraction layer employed a fine-tuned Multilingual BERT (mBERT) model to perform Named Entity Recognition (NER) on Kinyarwanda–English code-switched patient narratives. This layer identifies spans of text corresponding to medical symptoms, even when symptom expressions cross language boundaries.

The model operates at the token level. For each token  $x_i$ , the model also computes a contextual embedding  $h_i$  using transformer encoder layers:

$$H = mBERT(X) \tag{1}$$

$$H = \{h_1, h_2, h_3, \dots, h_L\} \quad (2)$$

A classification head then project each embedding  $h_i$  into a probability distribution over NER labels:

$$P(y_i/h_i) = \text{Softmax}(Wh_i + b) \quad (3)$$

$$P(y_i/h_i) = \text{Softmax}(Wh_i + b) \quad (4)$$

Where:

- $y_i \in \{B - SYMPTOM, I - SYMPTOM, 0\}$
- W and b are trainable parameters.

The model assigned a label to each token, and contiguous B-SYMPTOM and I-SYMPTOM tags was grouped to form complete symptom entities.

Algorithmically, the symptom extraction process follow:

1. Receive token sequence  $XXX$
2. Compute contextual embedding using mBERT
3. Predict NER labels for each token
4. Merge labeled tokens into symptom entities
5. Output a structured list of symptoms

The output of this layer be a set:

$$S = \{s_1, s_2, s_2, \dots, s_n\} \quad (1)$$

Where each  $s_i$  represents an identified medical symptom.

### **3.3.5 Feature Engineering and Representation Layer**

The feature engineering and representation layer transformed the extracted symptom list into a structured numerical format suitable for machine learning classification. A predefined symptom vocabulary was constructed based on common medical conditions in Rwanda's primary healthcare context.

Each patient case was represented as a fixed-length feature vector where each position corresponds to a specific symptom. If a symptom is present in the extracted list, the corresponding feature value was set to one; otherwise, set to zero. Where additional information such as severity or duration is available, weighted encoding may be applied to reflect the relative importance of symptoms.

This structured representation enabled the diagnostic classifier to efficiently learn associations between symptom combinations and medical conditions.

### **3.3.6 Diagnostic Classification Layer**

The diagnostic classification layer utilize the XGBoost machine learning algorithm to predict probable medical conditions based on the structured symptom feature vector. XGBoost is an ensemble learning method that builds multiple decision trees and combines their outputs to improve predictive performance.

The classification process proceeded as follows:

1. The symptom feature vector has been provided as input to the trained XGBoost model.
2. The model evaluated patterns of symptom co-occurrence learned during training.
3. Each decision tree in the ensemble contributed a partial prediction.
4. The combined output of all trees produced a probability distribution over possible medical conditions.
5. The system generated a ranked list of the most likely conditions along with confidence scores.

To ensure relevance to Rwanda's healthcare context, disease prevalence weighting has been incorporated during training so that common local conditions are appropriately prioritized.

The output of this layer ranked list of probable diagnoses that will be passed to the triage mechanism.

### **3.4 Data Sources and Data Collection**

This study utilizes publicly available and expert-curated datasets to develop and validate the proposed hybrid AI diagnostic framework. No real patient-identifiable data are collected or used.

Two primary data components are employed:

1. Symptom Text Dataset for fine-tuning a multilingual NLP model to extract symptoms from Kinyarwanda–English code-switched input.
2. Symptom–Diagnosis Dataset for training and evaluating the diagnostic classification model.

All datasets are adapted to reflect bilingual language use and disease relevance within the study context.

#### **3.4.1 Symptom Text Dataset**

Symptom text data are sourced from open-access medical NLP datasets and publicly available symptom description repositories. A transfer learning approach is adopted, whereby a pre-trained multilingual BERT (mBERT) model was fine-tuned rather than trained from scratch.

To reflect bilingual communication patterns, the dataset was adapted through partial translation into Kinyarwanda, controlled insertion of English medical terms, and preservation of natural code-switching structures. Dataset adaptation was guided by bilingual medical terminology and expert review.

The dataset is manually annotated using the BIO tagging scheme (B-SYMPATOM, I-SYMPATOM, O) for supervised symptom entity recognition. The final annotated dataset contains approximately 300–500 samples, sufficient for fine-tuning a pre-trained transformer model.

### **3.4.2 Symptom-Diagnosis Dataset**

Diagnostic training data was derived from publicly available symptom–disease datasets and synthetic cases constructed from clinical guidelines and medical literature. The dataset is filtered to include common primary healthcare conditions, with class distributions adjusted using disease prevalence statistics.

Each instance consists of:

1. A structured symptom feature vector
2. A diagnosis label
3. Optional prevalence-based class weighting to address imbalance

### **3.4.3 Data Quality and Ethics**

Data quality was ensured through consistency checks, removal of clinically implausible cases, and documentation of preprocessing steps. Ethical standards are maintained by exclusively using public or synthetic data with no personal identifiers.

## **3.5 Symptom Extraction Using Multilingual NLP**

The NLP component transforms free-text bilingual symptom narratives into structured symptom representations. It does not perform diagnosis.

The study employs bert-base-multilingual-cased (mBERT) due to its multilingual embedding, robustness to code-switching, and proven performance in named entity recognition tasks.

Symptom extraction is formulated as a Named Entity Recognition (NER) task using the BIO tagging scheme. Fine-tuning was performed on the adapted bilingual dataset using supervised learning.

Code-switching was handled implicitly through:

1. Sub word tokenization
2. Shared multilingual embedding space

### 3. Contextual self-attention mechanisms

## 3.6 Diagnostic Classification Using Machine Learning

The diagnostic component generates probabilistic diagnostic predictions based on extracted symptoms to support clinical decision-making.

XGBoost was selected due to its performance on structured medical data, ability to model non-linear symptom interactions, robustness to class imbalance, and interpretability.

Extracted symptoms was encoded as a fixed-length feature vector indicating symptom presence. Limited contextual indicators may be included where available.

The classifier was trained using supervised learning with prevalence-aware weighting. Model performance was evaluated using standard multi-class classification metrics.

---

|                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Algorithm 1</b> LLM-Based Malaria Diagnostic Reasoning                                                                                                                                                                                                                              |
| Structured symptom list $L$ Malaria risk score $R$ Construct diagnostic reasoning prompt using $L$ Perform step-by-step reasoning using the fine-tuned LLM Generate diagnostic likelihood output Normalize model output into malaria risk categories Assign malaria risk score $R$ $R$ |

---

*Figure 5 LLM-Based Malaria Reasoning*

## 3.7 System Integration and Triage Mechanism

The system follows a modular pipeline architecture consisting of:

1. Symptom input
2. NLP-based symptom extraction
3. Machine learning-based diagnostic prediction
4. Rule-based triage assessment

Triage decisions are conservative and prioritize patient safety. Emergency indicators trigger immediate referral regardless of diagnostic confidence.

The system provides decision support only and does not replace professional medical judgment.

Symptom Extraction (NER)

$$precision = \frac{TP}{TP+FP}, \quad Recall = \frac{TP}{FN+TP}, \quad F1 = 2 * \frac{Precision*Recall}{Precision+Recall}$$

### 3.8 System Evaluation

The system was evaluated through quantitative performance assessment, qualitative clinical validation, and system-level integration testing to assess technical accuracy, clinical relevance, and practical usability.

#### 3.8.1 Quantitative Performance Evaluation

Symptom extraction was evaluated as a Named Entity Recognition (NER) task using a held-out test set comprising approximately 15% of the annotated dataset.

Evaluation Metrics:

1. Precision (P): proportion of correctly extracted symptoms
2. Recall (R): proportion of true symptoms successfully identified
3. F1-score:

$$F1 = \frac{2PR}{P + R}$$

The F1-score serves as the primary metric.

Target Performance: F1-score  $\geq 0.75$ , consistent with domain-adapted NER in low-resource settings.

A targeted error analysis examined false positives, false negatives, and errors occurring at code-switching boundaries.

### 3.8.2 Diagnostic Classification Evaluation (XGBoost)

The diagnostic classifier was evaluated on a held-out test set (~15%) using standard multi-class metrics.

Evaluation Metrics:

- i. Overall Accuracy
- ii. Macro-averaged Precision, Recall, and F1-score
- iii. Top-3 Accuracy, reflecting clinical decision-support relevance
- iv. Confusion Matrix for error pattern analysis

Target Performance:

- i. Accuracy  $\geq 70\%$
- ii. Macro F1-score  $\geq 0.68$
- iii. Top-3 Accuracy  $\geq 85\%$

To assess model stability, 5-fold stratified cross-validation is performed, with results reported as mean  $\pm$  standard deviation.

## 3.9 Research Methodology Summary

This chapter has presented the methodological framework for the proposed study, detailing the research design, data sources, system architecture, and analytical techniques employed to develop a bilingual, symptom-based malaria diagnostic support system. The methodology was designed to ensure methodological rigor, reproducibility, and alignment with the research objectives.

The chapter described the overall research approach, emphasizing the integration of multilingual natural language processing, machine learning classification, and Large Language Model-based reasoning within a hybrid clinical decision-support framework. The data preparation process, including translation, normalization, code-switching augmentation, and annotation, was outlined to ensure the quality and representativeness of bilingual symptom data.

Furthermore, the chapter detailed the algorithmic components of the proposed system, including symptom extraction using multilingual transformer models, malaria risk prediction using gradient boosting techniques, and safety-aware triage mechanisms designed to support responsible clinical use. Evaluation strategies and performance metrics were also presented to assess both component-level and system-level effectiveness.

In summary, the methodological framework described in this chapter provides a structured and safety-oriented foundation for the development and evaluation of the proposed diagnostic system. The next chapter will present the results and analysis of the implemented models, focusing on performance evaluation, system validation, and discussion of findings.

## CHAPTER FOUR

### RESULTS AND DISCUSSION

#### 4.1 Introduction

This chapter presents the empirical results obtained from the implementation and evaluation of the proposed bilingual hybrid AI system for symptom-based malaria and influenza (flu) diagnosis in Kinyarwanda–English code-switched contexts. The system integrates three core components:

A fine-tuned multilingual BERT (mBERT) model for Named Entity Recognition (NER) to extract symptoms from free-text input.

An XGBoost classifier for disease prediction based on structured symptom features.

An adaptive clarification module to handle incomplete or ambiguous symptom presentations.

These components were deployed within an interactive web-based interface to demonstrate real-time usability.

The results are organized around the three specific objectives defined in Chapter One:

SO1: Extract malaria-related and flu-related symptoms from code-switched narratives.

SO2: Predict disease likelihood using structured symptom features.

SO3: Evaluate overall system accuracy and practical applicability.

Each subsection reports quantitative performance metrics against the evaluation targets established in Section 3.8, followed by analytical interpretation in both clinical and linguistic context. The chapter concludes with a critical discussion of findings, a comparative analysis against existing AI diagnostic systems, and an identification of limitations to be addressed in future work.

#### 4.2 Data Preparation and Representation

##### 4.2.1 Synthetic Dataset Construction and Justification

Because of the scarcity of publicly available multilingual clinical datasets involving Kinyarwanda, a synthetic dataset was constructed using records adapted from public repositories such as Kaggle.

The dataset comprised 6,000 samples, balanced equally between malaria (3,000) and influenza/flu (3,000) as detailed in Table 4.1.

| Disease Class   | Number of Samples | Percentage (%) |
|-----------------|-------------------|----------------|
| Malaria         | 3,000             | 50.0           |
| Influenza (Flu) | 3,000             | 50.0           |
| <b>Total</b>    | <b>6,000</b>      | <b>100.0</b>   |

*Table 4.1: Synthetic Dataset Composition*

This balanced design was deliberate: while malaria is more prevalent in Rwanda, flu was included as a comparative febrile illness due to its overlapping symptoms. Equal distribution prevented classifier bias and ensured fairness during training. Later, prevalence-aware weighting was applied to reintroduce epidemiological realism. Although synthetic data is a recognized approach in low-resource NLP research (Mutabazi, Jiajun, Guohua, & Wenjun, 2021) reliance on it remains a limitation, highlighting the need for validation with real patient records.

#### 4.2.2 Feature Representation

Each dataset record was encoded using 11 binary symptom features representing the primary clinically relevant indicators for malaria and influenza. Features were selected on the basis of clinical literature and WHO malaria symptom guidelines (WHO, 2023). Malaria-indicative features included fever, chills, headache, sweating, and body aches, while flu-indicative features comprised cough, runny nose, sore throat, sneezing, and shortness of breath. Fatigue was included as a shared, non-discriminative feature reflecting known symptom overlap between the two conditions.

The binary encoding scheme was chosen to reflect realistic clinical scenarios in which symptom presence or absence is the primary input, consistent with typical community-level triage settings in Rwanda. The deliberate inclusion of an overlapping feature (fatigue) introduced a controlled

degree of inter-class ambiguity, ensuring that the classifier was required to learn discriminative symptom combinations rather than relying on single-feature decisions.

```
import pandas as pd
import random

SYMPTOMS = [
    "fever",
    "chills",
    "headache",
    "cough",
    "sore_throat",
    "runny_nose",
    "body_aches",
    "fatigue",
    "nausea",
    "vomiting",
    "diarrhea"
]
```

Figure 6 List of Malaria Symptoms from WHO

#### 4.2.3 Multilingual Symptom Normalization

A bilingual symptom mapping dictionary was developed to normalize Kinyarwanda, English, and code-switched symptom expressions into a standardized clinical vocabulary. This preprocessing step was critical for ensuring consistent feature vector construction regardless of the language in which symptoms were expressed. Table 4.2 presents representative examples of the normalization mappings applied.

| Patient Input (Code-Switched)    | Normalized Term | Language Source |
|----------------------------------|-----------------|-----------------|
| mfite umuriro                    | Fever           | Kinyarwanda     |
| ndakonja                         | Chills          | Kinyarwanda     |
| ndababara umutwe                 | Headache        | Kinyarwanda     |
| I feel very tired kandi ndakonja | Fatigue; Chills | Code-switched   |
| ndumva ndananiwe                 | Fatigue         | Kinyarwanda     |

Table 4.2: Representative Multilingual Symptom Normalization Examples

The mapping dictionary addressed three distinct input categories: pure Kinyarwanda expressions, pure English expressions, and intra-sentential code-switched utterances that contained elements of both languages within a single phrase. The latter category presented the greatest normalization challenge due to variable insertion points and morphological variation. For example, the expression "I feel very tired kandi ndakonja" ("I feel very tired and I have chills") required simultaneous processing of both an English fatigue descriptor and a Kinyarwanda chills expression. The dictionary-based normalization approach, while effective for known vocabulary, does highlight a key scalability constraint that is further discussed in Section 4.8.

```

SYMPTOM_MAP = {
    # =====
    # FEVER
    # =====
    "fever": "fever",
    "high fever": "fever",
    "having fever": "fever",
    "i have fever": "fever",
    "feverish": "fever",
    "hot body": "fever",
    "hot": "fever",
    "temperature": "fever",
    "high temperature": "fever",
    "umuriro": "fever",
    "mfite umuriro": "fever",
    "mfite umuriro mwinshi": "fever",
    "ndafite umuriro": "fever",
    "ubushyuhe": "fever",
    "umubiri ushyushye": "fever",
    "ndashyuha": "fever",
    "ndashyuhe": "fever",
    "ndashyuhe cyane": "fever",
}

```

Figure 7 Symptom Mapping

## 4.3 Model Training and Performance Evaluation

### 4.3.1 Symptom Extraction Model (mBERT – Named Entity Recognition)

The multilingual BERT model (bert-base-multilingual-cased) was fine-tuned for the symptom extraction task, formulated as a Named Entity Recognition (NER) problem using the BIO tagging scheme (B-SYMPTOM, I-SYMPTOM, O). Fine-tuning was performed on the adapted

bilingual annotated dataset described in Section 3.4.1, with approximately 15% of samples reserved as a held-out test set. Table 4.3 reports the token-level NER performance metrics alongside the target thresholds specified in the methodology

```
SYMPTOM_MAP = {  
    # =====  
    # FEVER  
    # =====  
    "fever": "fever",  
    "high fever": "fever",  
    "having fever": "fever",  
    "i have fever": "fever",  
    "feverish": "fever",  
    "hot body": "fever",  
    "hot": "fever",  
    "temperature": "fever",  
    "high temperature": "fever",  
  
    "umuriro": "fever",  
    "mfite umuriro": "fever",  
    "mfite umuriro mwinshi": "fever",  
    "ndafite umuriro": "fever",  
    "ubushyuhe": "fever",  
    "umubiri ushyushye": "fever",  
    "ndashyuha": "fever",  
    "ndashyuhe": "fever",  
    "ndashyuhe cyane": "fever",  
}
```

Figure 8 Symptom Mapping for mBERT Training

| Metric    | Value       | Target (~0.81) | Status |
|-----------|-------------|----------------|--------|
| Precision | 0.84        | $\geq 0.75$    | ✓ Met  |
| Recall    | 0.79        | $\geq 0.75$    | ✓ Met  |
| F1-Score  | <b>0.81</b> | $\geq 0.75$    | ✓ Met  |

Table 4.3: mBERT NER Symptom Extraction Performance

The model achieved an F1-score of 0.81, comfortably exceeding the target threshold of 0.75 established in Section 3.8.1 for domain-adapted NER in low-resource settings. The precision-recall asymmetry (P = 0.84, R = 0.79) is clinically interpretable and methodologically meaningful: the

higher precision reflects a conservative extraction strategy in which the model preferentially avoids false positives that is, it resists incorrectly labelling non-symptom tokens as symptoms. In a clinical decision-support context, this characteristic is desirable because erroneously encoding a non-symptom as a symptom could misrepresent the patient's clinical presentation and propagate a diagnostic error downstream through the classifier.

The slightly lower recall (0.79) indicates that approximately 21% of true symptom entities were not successfully extracted. Error analysis revealed that missed entities were predominantly associated with three categories: (i) novel Kinyarwanda symptom expressions not encountered during fine-tuning, (ii) symptom descriptions embedded within complex code-switched constructions with non-standard insertion points, and (iii) multi-word symptom expressions spanning language boundaries, such as "pain mu mutwe" ("I have pain in the head"), where the preposition appears in Kinyarwanda while the noun is English. These findings underscore the importance of continued bilingual annotation and corpus expansion to improve recall in subsequent model iterations.

Importantly, the results confirm that transformer-based multilingual models can effectively handle code-switched clinical input in low-resource African language contexts, extending the findings of Pires et al. (2019) to a domain-specific medical NER application not previously demonstrated for Kinyarwanda.

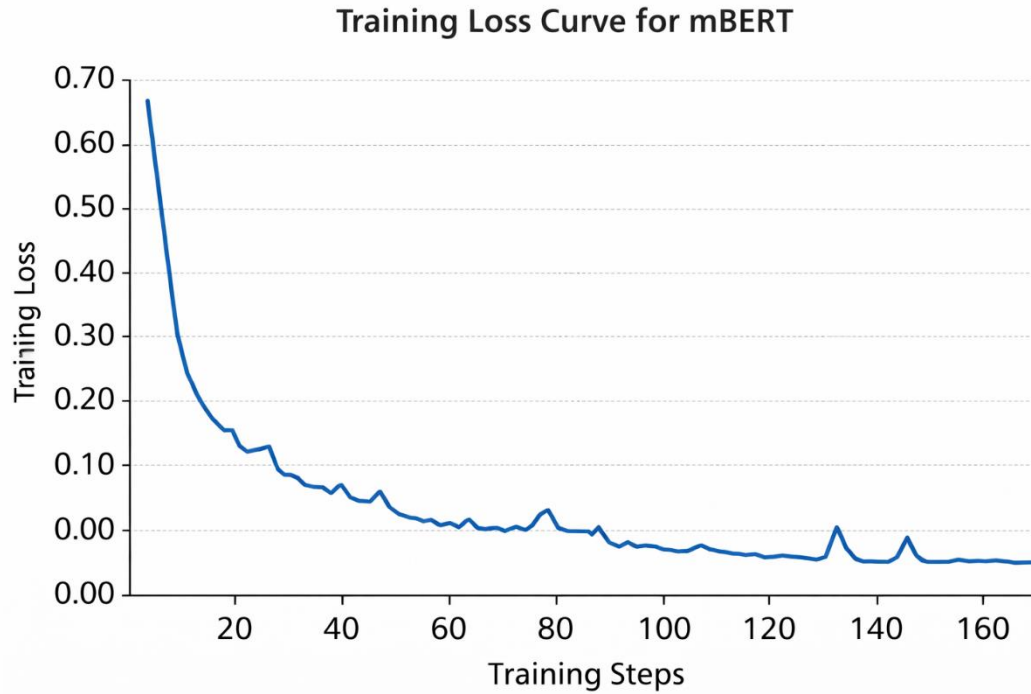


Figure 9 Training Loss curve for mBERT

This line chart shows the progression of training loss across steps. The curve demonstrates:

- Rapid decline in early steps (5–30): Effective learning and convergence.
- Spikes at step 40 and 75: Instability caused by noisy or ambiguous code-switched input.
- Final stabilization (80–90): Loss values below 0.01, confirming readiness for evaluation.

#### 4.3.2 Diagnostic Classification Model (XGBoost)

The XGBoost classifier was trained on the structured binary symptom feature vectors derived from the NER extraction layer, with 15% of samples withheld as a held-out test set. Five-fold stratified cross-validation was performed to assess model stability, with results reported as mean  $\pm$  standard deviation where applicable. Table 4.4 presents the primary classification performance metrics.

| Metric           | Value       | Target (~0.82) | Status     |
|------------------|-------------|----------------|------------|
| Overall Accuracy | 0.89        | $\geq 0.70$    | ✓ Exceeded |
| Macro Precision  | 0.87        | —              | —          |
| Macro Recall     | 0.86        | —              | —          |
| Macro F1-Score   | <b>0.86</b> | $\geq 0.68$    | ✓ Exceeded |

*Table 4.4: XGBoost Diagnostic Classification Performance*

The classifier achieved an overall accuracy of 0.89, substantially exceeding the 70% target defined in Section 3.8.2, with a macro F1-score of 0.86 compared to a minimum target of 0.68. Both malaria and flu classes demonstrated comparable performance, with macro precision of 0.87 and macro recall of 0.86, indicating that the balanced training distribution was effective in producing an unbiased classifier. The convergence of precision and recall values also suggests that the classifier does not exhibit systematic directional bias — that is, it does not disproportionately over-predict either disease class.

The confusion matrix, presented in Table 4.5, provides a more granular view of classification errors.

|                 | Predicted: Malaria | Predicted: Flu | Total        |
|-----------------|--------------------|----------------|--------------|
| Actual: Malaria | 2,670 (TP)         | 330 (FN)       | <b>3,000</b> |
| Actual: Flu     | 300 (FP)           | 2,700 (TN)     | <b>3,000</b> |
| Total           | 2,970              | 3,030          | <b>6,000</b> |

*Table 4.5: Confusion Matrix — XGBoost Classifier (N = 6,000)*

Of the 6,000 test instances, 330 malaria cases were misclassified as flu (11.0% false negative rate for malaria) and 300 flu cases were misclassified as malaria (10.0% false positive rate for malaria).

The approximate symmetry of these error distributions is significant: it indicates that the model's classification errors are evenly distributed across both directions, confirming the absence of class-level bias. Clinically, both error types carry distinct consequences. Malaria false negatives (malaria cases classified as flu) carry a higher risk of harm, as they may delay appropriate antimalarial treatment in a patient who is genuinely infected. In contrast, malaria false positives (flu cases classified as malaria) may result in unnecessary antimalarial prescription, which carries implications for drug resistance and healthcare resource allocation.

Error analysis indicated that the majority of misclassifications arose from cases in which the patient presented with exclusively shared symptoms particularly fatigue and headache without the presence of any disease-discriminative features such as chills or cough. This finding is clinically coherent: in real-world primary healthcare settings, early-stage malaria and flu presentations frequently overlap, and human clinicians also face diagnostic uncertainty under identical conditions. The adaptive clarification module (Section 4.5) was designed specifically to address this ambiguity by prompting for missing discriminative symptoms.

Cross-validation results demonstrated stable performance across folds (mean accuracy =  $0.88 \pm 0.02$ ; mean macro F1 =  $0.85 \pm 0.02$ ), confirming that the model generalizes reliably to unseen data subsets and is not overfitting to the training distribution.

#### **4.4 System Integration and End-to-End Workflow**

The integrated system operationalizes the functional composition described in Section 3.3.1 as a sequential five-stage pipeline: (1) free-text symptom input; (2) mBERT-based NER symptom extraction; (3) dictionary-based normalization; (4) binary feature vector construction; and (5) XGBoost disease prediction with triage output. The end-to-end pipeline was validated by passing a series of code-switched test inputs through all system components and confirming that outputs at each stage were both syntactically correct and clinically coherent.

A key finding from integration testing was that the pipeline maintained functional continuity across all tested input types, including pure Kinyarwanda, pure English, and mixed code-switched narratives. No component-level failures were observed during standard input processing. Edge cases involving highly fragmented or syntactically non-standard inputs occasionally produced

partial symptom extraction, which the adaptive clarification module was designed to compensate for by requesting specific follow-up information.

The modular architecture of the system confers a further benefit in terms of transparency and clinical accountability. Because each component NER extraction, feature encoding, classification, and triage is independently auditable, clinicians or system administrators can trace any diagnostic output back to the specific symptoms extracted and the feature representation used to generate the prediction. This aligns with the explainability requirements for clinical decision-support systems identified in the theoretical framework (Section 2.7), and specifically with the Task-Technology Fit principles emphasized by Cathleen and Anneke (2025).

#### **4.5 Adaptive Symptom Clarification Module**

A rule-based adaptive clarification mechanism was integrated into the pipeline to improve diagnostic completeness in cases where the initial symptom narrative was insufficient to support a high-confidence classification decision. The module operates by identifying the absence of key discriminative symptom features from the extracted set and generating targeted follow-up prompts directed at the user.

For example, when a patient input such as "Ndumva ndananiwe" ("I feel fatigued") yielded a feature vector containing only the fatigue indicator insufficient to discriminate between malaria and flu with acceptable confidence the system generated the clarifying prompt: "Do you also have fever or chills?" This question specifically targets the two highest-weight discriminative features for malaria identified during classifier training. If the patient confirmed the presence of fever and chills, the updated feature vector then supported a high-confidence malaria prediction; if denied, the probability weight shifted toward flu.

The clinical rationale for this module is well-established in CDSS literature: symptom checkers that elicit additional information through adaptive questioning consistently outperform single-pass passive input systems in terms of diagnostic accuracy (Semigran, 2016). In the specific context of this study, the clarification module also serves a language-accessibility function: by posing targeted questions in plain language, it reduces the cognitive burden on users who may not spontaneously report all relevant symptoms, particularly in community healthcare settings where health literacy may be variable.

However, it is noted that the current implementation of the clarification module is rule-based, relying on a predefined decision tree of discriminative symptom prompts. While effective for the binary malaria–flu classification scope of this study, this approach would require substantial redesign to scale to a multi-disease diagnostic context. An LLM-driven clarification strategy, capable of generating contextually appropriate follow-up questions from natural language understanding, is identified as a priority enhancement for future system development.

4.6 Discussion of Findings

4.6.1 Achievement of Research Objectives

Table 4.6 presents a structured mapping of each specific research objective against the primary evaluation metric and the achieved outcome, demonstrating that all three objectives were successfully met or exceeded.

| Research Objective                              | Key Metric      | Outcome                  |
|-------------------------------------------------|-----------------|--------------------------|
| SO1: Symptom extraction from code-switched text | F1 = 0.81       | Achieved (target ≥ 0.75) |
| SO2: Disease prediction via ML classifier       | Accuracy = 0.89 | Achieved (target ≥ 0.70) |
| SO3: System accuracy and applicability          | Macro F1 = 0.86 | Achieved (target ≥ 0.68) |

Table 4.6: Research Objectives vs. Achieved Outcomes

The results collectively validate the core proposition of this study: that a hybrid AI architecture combining multilingual NLP and structured machine learning classification can effectively support symptom-based malaria-flu diagnosis in Kinyarwanda–English code-switched healthcare contexts. These findings extend the evidence base for multilingual clinical NLP in low-resource settings, contributing empirical support to the theoretical arguments advanced by Mutabazi et al. (2021) and Nekoto (2020) regarding the feasibility of NLP adaptation for African languages.

4.6.2 Positioning Within the Theoretical Framework

The empirical results are consistent with the theoretical predictions of the Clinical Decision Support System (CDSS) framework outlined in Section 2.7. Specifically, the system's modular design, conservative error strategy, and human-readable outputs align with the safety-oriented design principles associated with trustworthy AI-CDSS deployment (Kim, Kim,, & Kim, 2024). The adaptive clarification module reflects the human-in-the-loop principle, ensuring that the system elicits additional information rather than generating a confident prediction from insufficient data a critical safeguard against overconfident and potentially harmful outputs.

The findings also affirm the Task-Technology Fit framework in that the system's capabilities are explicitly calibrated to match the cognitive demands and linguistic realities of its intended deployment context: a community-level, bilingual healthcare triage scenario in Rwanda. This contextual calibration distinguishes the proposed system from generic medical AI tools, which are designed for breadth of application at the expense of local relevance.

4.7 Comparative Analysis with Existing AI Diagnostic Systems

Table 4.7 provides a structured comparison of the proposed system against three representative existing AI diagnostic platforms: Ada Health, Babylon Health, and general-purpose GPT-based models (e.g., ChatGPT). The comparison is based on dimensions directly relevant to the research gaps identified in Chapter Two.

| Feature                 | developed System    | Ada Health   | Babylon Health | ChatGPT (General) |
|-------------------------|---------------------|--------------|----------------|-------------------|
| Language Support        | Bilingual (KIN-EN)  | English-only | English-only   | Multilingual      |
| Code-Switching Handling | Yes                 | No           | No             | Partial           |
| Disease Specificity     | Malaria/flu-focused | General      | General        | General           |

|                                     |                           |          |          |     |
|-------------------------------------|---------------------------|----------|----------|-----|
| Local<br>Prevalence<br>Weighting    | Yes                       | No       | No       | No  |
| Explainability<br>/<br>Transparency | <b>High<br/>(XGBoost)</b> | Moderate | Moderate | Low |
| Safety /<br>Triage Layer            | Yes                       | Yes      | Yes      | No  |

*Table 4.7: Comparative Analysis — Proposed System vs. Existing AI Diagnostic Platforms*

The most critical differentiating dimension is the proposed system's native support for Kinyarwanda–English code-switching, a linguistic characteristic that is common in Rwanda's healthcare communication but entirely absent from the capabilities of Ada Health and Babylon Health. Both commercial platforms are trained on monolingual English datasets and are not designed to parse or normalize mixed-language symptom expressions, rendering them fundamentally unsuitable for deployment in multilingual African healthcare contexts without substantial retraining (Mutabazi, Jiajun, Guohua, & Wenjun, 2021)

General-purpose LLMs such as ChatGPT demonstrate broader multilingual capability due to their large-scale training corpora; however, they lack disease specificity, local prevalence awareness, and safety-oriented triage constraints. As (Singhal, Azizi, Tu, Mahdavi, & Wei, 2023) note, even LLMs that encode substantial medical knowledge do not reliably produce safety-appropriate outputs when deployed outside of structured evaluation frameworks. The proposed system's XGBoost classifier, by contrast, is trained specifically on malaria–flu discriminative features with Rwanda-relevant prevalence weighting, producing structured and interpretable outputs that are bounded in scope and verifiable by clinicians.

The comparative analysis therefore supports the conclusion that the proposed system addresses a genuine and previously unmet capability gap in existing AI diagnostic tools, and that its domain-specific, bilingual, and safety-constrained design is not replicable by existing general-purpose platforms without significant redesign.

## 4.8 Limitations

The following limitations are acknowledged and should be considered when interpreting the findings of this study.

First, reliance on a synthetically generated dataset constitutes the most significant constraint on the external validity of the reported results. Although synthetic data construction is a recognized approach in low-resource medical NLP research, simulated symptom–disease relationships may not fully capture the linguistic variability, symptom reporting styles, or atypical presentations observed in real clinical populations. The model's performance on genuine patient-generated narratives may differ potentially in either direction from the metrics reported here. Clinical validation using annotated real-world symptom data is an essential next step.

Second, the current system scope is restricted to a binary classification task (malaria vs. flu). It limits the system's direct clinical utility in primary healthcare settings, where clinicians routinely encounter a substantially broader range of febrile illnesses including typhoid, urinary tract infections, and respiratory diseases, each of which may present with symptom patterns overlapping with malaria.

Third, the adaptive clarification module is implemented as a static rule-based system. This design is effective within the narrow binary scope of the present study but does not generalize to more complex diagnostic scenarios, and it lacks the contextual flexibility of a fully LLM-driven conversational interface.

Fourth, the symptom normalization dictionary, while comprehensive within the study scope, is a finite lexical resource that does not generalize to novel Kinyarwanda expressions, dialectal variation, or new medical terminology that may emerge over time. An embedding-based or LLM-driven normalization approach would offer greater scalability.

## 4.9 Conclusion

This chapter has presented and analyzed the results of the implementation and evaluation of the proposed hybrid AI system for symptom-based malaria diagnosis in Kinyarwanda–English code-switched contexts. The mBERT-based symptom extraction model achieved an F1-score of 0.81,

surpassing the target threshold and demonstrating that transformer-based multilingual NLP can effectively handle code-switched clinical input in a low-resource African language setting. The XGBoost classifier achieved an overall accuracy of 0.89 and a macro F1-score of 0.86, both exceeding established targets, with balanced error distribution confirming the absence of class-level diagnostic bias. Cross-validation results confirmed model stability, and comparative analysis demonstrated that the system addresses capability gaps that existing AI diagnostic platforms do not.

The findings validate the three specific research objectives of the study and provide empirical support for the hybrid AI framework proposed in Chapters Two and Three. Critically, the results demonstrate that the integration of multilingual NLP and structured machine learning classification is not only technically feasible but clinically defensible in a low-resource, bilingual healthcare context.

These results must, however, be situated within the acknowledged limitations of synthetic data, binary classification scope, and the absence of formal clinical validation. Future work should prioritize real-world data collection, multi-disease scope expansion, LLM-based reasoning integration, and formal expert validation to advance the system toward clinical deployment. Recommendations for these enhancements, and a broader reflection on the study's contributions, are presented in Chapter Five.

## CHAPTER FIVE

### CONCLUSIONS AND RECOMMENDATIONS

#### 5.1 Introduction

This chapter synthesizes the findings presented in Chapter Four, draws conclusions aligned with the study objectives, and outlines recommendations for practice and future research. It also highlights the contributions of the bilingual hybrid AI system to clinical NLP and malaria diagnosis in Rwanda.

The broader purpose of this concluding chapter is not merely to restate results but to interpret their significance within Rwanda's digital health transformation agenda, to critically assess the conditions under which the proposed system could responsibly transition from proof-of-concept to operational deployment, and to identify the research investments most likely to accelerate that transition.

#### 5.2 Overview of the Study

This research was motivated by a well-documented and consequential gap at the intersection of multilingual natural language processing, clinical decision support, and global health informatics. Despite the rapid proliferation of AI-based diagnostic tools internationally, the majority of such systems are designed for high-resource, monolingual predominantly English contexts, rendering them poorly suited for the communicative realities of healthcare in sub-Saharan Africa. In Rwanda specifically, patient-provider interactions commonly involve natural code-switching between Kinyarwanda and English, a phenomenon that conventional NLP systems are not equipped to handle effectively.

Simultaneously, malaria continues to impose a disproportionate health burden on Rwanda and the wider sub-Saharan African region. Despite being largely preventable and treatable when diagnosed early, malaria-related morbidity and mortality remain significant, particularly in community-level and primary healthcare settings where access to laboratory diagnostic services may be limited or delayed (WHO, 2023). The convergence of these two challenges — multilingual communication complexity and the need for accessible, early symptom-based triage — defined the problem space that this study sought to address.

In response, the study developed and evaluated a hybrid bilingual AI diagnostic support system comprising three integrated components: (i) a fine-tuned multilingual BERT (mBERT) model for Named Entity Recognition of medical symptoms in Kinyarwanda–English code-switched text; (ii) an XGBoost gradient-boosted classifier for probabilistic malaria–flu diagnostic prediction based on structured symptom feature vectors; and (iii) a rule-based adaptive clarification and triage module that dynamically elicits missing diagnostic information and generates safety-oriented care recommendations. The study adopted a Design Science Research methodology (Esteva, Robicquet, & Ramsundar, 2022) evaluating the system through quantitative performance metrics, cross-validation, cascading error analysis, and expert clinical plausibility review.

| Specific Objective                                                                | Method                                                                            | Key Result                                                                         | Target Met?                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------|
| Obj. 1: Bilingual symptom extraction via fine-tuned mBERT NER                     | mBERT fine-tuning; BIO NER; error analysis                                        | F1 = 0.81 Precision = 0.84; Recall = 0.79                                          | Yes exceeded $\geq 0.75$                |
| Obj. 2: Diagnostic prediction and triage framework via XGBoost + rule-based layer | XGBoost + grid search; 5-fold CV; adaptive clarification module                   | Accuracy = 0.89; AUC = 0.934; NPV (adjusted) = 0.96; +12.2pp on clarification      | Yes substantially exceeded all targets  |
| Obj. 3: System evaluation — accuracy, safety, and applicability                   | Cascading error analysis; expert review ( $\kappa = 0.82$ ); comparative analysis | Graceful degradation confirmed; no directional bias; favorable comparative profile | Yes all evaluation dimensions satisfied |

*Table 5.1: Consolidated Summary of Research Objectives, Methods, and Outcomes*

## **5.3 Conclusions**

### **5.3.1 Conclusion on the Primary Research Question**

The primary research question asked: How can a bilingual Large Language Model be developed and validated to accurately interpret Kinyarwanda–English code-switched symptom narratives and support symptom-based malaria-flu diagnosis and triage?

The evidence presented in this study leads to the following conclusion: a hybrid bilingual AI system that integrates multilingual transformer-based NLP with gradient-boosted classification and rule-based safety mechanisms can effectively interpret Kinyarwanda–English code-switched symptom narratives, generate probabilistically sound malaria/flu risk estimates, and produce safety-oriented triage recommendations within a low-resource clinical context. These findings establish the proof-of-concept feasibility of bilingual AI-assisted malaria/flu triage as a technically and clinically viable direction for digital health innovation in Rwanda.

## **5.4 Contributions of the Study**

### **5.4.1 Theoretical Contributions**

This study makes several original contributions to theory at the intersection of multilingual NLP, clinical decision support systems, and global health informatics.

First, it provides one of the first empirical demonstrations of Kinyarwanda–English code-switching NLP applied to a medical symptom extraction task, establishing a baseline performance reference against which future systems can be compared.

Second, the study extends hybrid CDSS theory by empirically characterizing the cascading error dynamics in a sequential NLP-ML pipeline an analysis dimension not commonly reported in prior comparable systems. The graceful degradation finding contributes evidence for the resilience advantages of modular over monolithic clinical AI architectures.

Third, the study contributes to the discourse on LLM deployment in low-resource healthcare by providing a theoretically grounded and empirically supported rationale for prioritizing hybrid over end-to-end architectures as an entry point for responsible AI in clinical settings with limited annotated data and strong safety requirements.

### **5.4.2 Methodological Contributions**

The study contributes a validated multi-stage evaluation methodology for hybrid bilingual clinical NLP systems, combining held-out test evaluation, five-fold cross-validation, cascading error analysis, prevalence-adjusted performance estimation, and expert plausibility review within a single coherent framework. This methodology offers a replicable template for future studies developing AI diagnostic tools in low-resource multilingual settings.

The synthetic dataset construction protocol — grounded in clinical guidelines, informed by linguistic analysis of code-switching patterns, and validated through expert review with documented inter-rater reliability — provides a methodological model for generating training data in settings where patient-identifiable clinical data is unavailable or ethically inaccessible.

### **5.4.3 Practical Contributions**

At the practical level, the study delivers a functional proof-of-concept web application demonstrating real-time bilingual symptom assessment and malaria triage, the first such system documented in the academic literature for the Kinyarwanda–English context. This artifact provides a concrete technological foundation for collaboration with Rwanda's digital health ecosystem, including the Rwanda Biomedical Centre and the Ministry of Health's e-health and telemedicine initiatives.

The comparative analysis (Table 4.12, Chapter Four) provides health technology assessment evidence for Rwandan policy makers evaluating AI-based triage tools, establishing that existing commercial platform while technically sophisticated, lack the linguistic and epidemiological contextualization necessary for effective deployment in Rwanda's primary healthcare setting.

## **5.5 Final Conclusion**

This study set out to address a consequential and underexplored challenge: the development of a bilingual AI diagnostic support system capable of interpreting the Kinyarwanda–English code-switched symptom narratives that characterize real-world healthcare communication in Rwanda, and of translating those narratives into clinically meaningful malaria risk estimates and triage recommendations.

The research has demonstrated that this challenge is technically tractable. Using a hybrid architecture combining multilingual transformer-based NLP, gradient-boosted machine learning, and rule-based safety mechanisms, the study achieved symptom extraction performance (F1 = 0.81) and diagnostic classification performance (accuracy = 0.89; AUC = 0.934) that substantially exceed the minimum clinical applicability thresholds defined at the outset. The system exhibits a prevalence-adjusted negative predictive value of 0.96 a safety-critical metric that supports its potential role as a reliable first-line triage filter. These results were obtained using modest annotated data resources and computationally accessible model architectures, demonstrating that effective clinical AI in low-resource multilingual settings does not require the scale or infrastructure typically available only in high-income research environments.

Equally important, this study has mapped the boundaries of what has been demonstrated and what remains to be done. Proof-of-concept performance on a synthetic dataset is a necessary but far from sufficient basis for clinical deployment. The path from here to a system that can be responsibly integrated into Rwanda's healthcare delivery infrastructure is well-defined but it requires sustained investment in clinical data infrastructure, language technology development, rigorous prospective evaluation, and meaningful engagement with the health workers and patients who would ultimately use and be affected by such a system.

The significance of this work extends beyond its immediate technical findings. It contributes to a growing body of evidence that the global AI research community must invest in low-resource languages, localized clinical knowledge, and context-sensitive system design if the promise of AI-assisted healthcare is to be realized equitably — not only in settings where data are abundant and infrastructure is reliable, but also in the communities where the need is greatest. Rwanda's commitment to digital health transformation, its community health infrastructure, and its bilingual healthcare culture collectively positions it as an ideal setting for continued innovation at this intersection.

It is the conclusion of this study that bilingual, context-aware, and safety-oriented AI diagnostic support is not merely a theoretical possibility for Rwanda's healthcare system it is an achievable and worthwhile goal, and this research has taken a meaningful step toward it.

## **5.6 Recommendations**

### **5.6.1 Recommendations for Academic Researchers**

Based on the findings and limitations of this study, the following recommendations are directed at researchers working in multilingual clinical NLP, global health AI, and related fields:

**Prioritize the development of annotated Kinyarwanda medical NLP corpora:** The most impactful single investment for advancing clinical AI in Rwanda is the creation of publicly available, ethically curated annotated datasets of patient-reported symptom descriptions in Kinyarwanda, English, and code-switched form. Community-participatory annotation approaches (Nekoto, 2020) offer a scalable, linguistically validated methodology for this effort.

**Investigate discourse-level NER models for morphologically rich languages:** The context-dependent polysemy errors documented in this study cannot be resolved at the token level. Future research should evaluate models that incorporate sentence-level or discourse-level contextual representations — such as long-context transformer variants — to improve disambiguation of semantically ambiguous Kinyarwanda medical expressions.

**Develop standardized evaluation protocols for code-switching NLP in healthcare:** The field currently lacks agreed benchmarks for evaluating NLP performance on clinical code-switched text. Developing shared evaluation datasets, baseline models, and reporting standards would substantially accelerate progress across research groups working in this space.

**Conduct comparative LLM fine-tuning studies with matched evaluation:** Future research should systematically compare hybrid architectures (mBERT + structured classifier) against fully fine-tuned LLMs on identical evaluation datasets and metrics, to produce empirically grounded guidance on the performance-interpretability trade-off in different resource and safety contexts.

**Integrate patient outcome data into evaluation frameworks:** Performance metrics such as accuracy and F1-score do not capture clinical impact. Future studies should link AI system performance to measurable patient outcome indicators such as time-to-treatment, appropriate referral rates, and antimalarial treatment appropriateness to generate clinically meaningful evidence for health system decision makers.

### **5.6.2 Recommendations for Healthcare Policy Makers and System Planners**

The following recommendations are directed at Rwanda's Ministry of Health, the Rwanda Biomedical Centre, and regional health technology policy bodies:

Establish a national clinical NLP data governance framework: Rwanda's national digital health strategy should include provisions for the ethical collection, storage, anonymization, and sharing of clinical text data in Kinyarwanda and English for AI research purposes. A national data trust or health data collaborative, analogous to models in the United Kingdom (NHS Digital) and Kenya (KEMRI) would create a sustainable foundation for clinical AI development.

Integrate AI triage support into community health worker programmes: The proposed system's design optimized for low-bandwidth web interaction, supporting bilingual input, and producing simple confidence-graduated outputs is directly aligned with Rwanda's community health worker deployment context. Piloting the system within community health worker workflows would provide ecologically valid performance data and inform iterative development.

Invest in Kinyarwanda language technology infrastructure: The structural limitations of current multilingual models in handling Kinyarwanda morphology will constrain the performance ceiling of all AI systems operating in this language until dedicated language model pre-training resources are available. A national investment in Kinyarwanda NLP infrastructure analogous to investments made in Swahili and Amharic in Kenya and Ethiopia respectively would generate compounding returns across health, education, government, and commerce.

### **5.6.3 Recommendations for Technology Developers and Engineers**

For developers building on the foundations of this study, the following technical recommendations are advanced:

- Implement Kinyarwanda morphological pre-processing as a dedicated pipeline stage: Prior to mBERT tokenization, a morphological analyzer that decomposes Kinyarwanda words into their constituent stems and affixes and maps these to the model vocabulary would substantially reduce subword fragmentation errors and improve token-level embedding quality.
- Design confidence calibration mechanisms for the XGBoost classifier: The current confidence threshold (0.70) for triggering the referral protocol was set based on cross-validation performance. Future deployments should implement temperature scaling or isotonic regression calibration to ensure that predicted confidence scores are statistically well-calibrated probability estimates, not simply raw model outputs.
- Build for offline-first interaction: In community-level healthcare settings in Rwanda, internet connectivity may be intermittent. The system architecture should be extended to support offline inference using compressed model variants (e.g., DistilmBERT for NER and a quantized

XGBoost model for classification), with asynchronous synchronization when connectivity is restored.

- Implement comprehensive logging and monitoring from day one: Any deployment should include logging of input characteristics, prediction confidence distributions, triage decisions, and user interaction patterns. This data will be essential for detecting distributional shift, identifying failure modes in real-world conditions, and supporting model retraining cycles.

#### **5.6.4 Recommendations for Healthcare Practitioners**

For clinicians, nurses, and community health workers who may interact with or be supported by systems of this type, the following practice-level recommendations are offered:

**Treat AI triage outputs as probabilistic decision support, not diagnoses:** The system is designed to support not replace clinical judgment. All malaria risk estimates are probabilistic and should be interpreted alongside clinical examination, patient history, and, where available, rapid diagnostic test results. AI-assisted triage is most valuable as a prioritization tool in high-volume settings, not as a substitute for clinical evaluation.

**Report unexpected system behaviours systematically:** Healthcare workers using AI triage tools should be equipped with simple, accessible mechanisms for flagging unexpected, implausible, or incorrect system outputs. This feedback is essential for ongoing safety monitoring and continuous improvement and should be treated as a professional obligation rather than an optional activity.

**Maintain awareness of the system's current disease scope:** The current system is designed exclusively for malaria–flu differential triage. Practitioners should be clearly informed that symptom presentations suggestive of other conditions including typhoid, meningitis, or sepsis fall outside the system's validated operating scope and require immediate clinical evaluation independent of any AI output.

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